30 FUTURES PLATFORM THEMES

Deep Academic Analysis: Sources, Focus Areas & Strategic Recommendations

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Prepared from the perspective of Futures Studies Academic Analysis

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EXECUTIVE SUMMARY

This report provides a comprehensive academic evaluation of the 30 Futures Platform Themes framework. For each of the 30 themes, the analysis: (1) critically evaluates the current focus area description; (2) proposes a revised, futures-oriented description with argumentation; (3) audits the existing source lists for quality and completeness; and (4) delivers a final, curated list of 12–15 authoritative sources per theme.

The analysis is conducted from the standpoint of an academic professional in futures studies, applying standards of source authority, methodological rigour, institutional independence, and thematic relevance.

Key Findings

STRUCTURAL OBSERVATIONS

- The six-category framework is logically coherent. However, three significant thematic overlaps require attention: Theme 2 (Data & Digitalisation) vs. Theme 7 (Digital & Sharing Economies) both reference "platforms"; Theme 10 (Construction & Urbanisation) and Theme 30 (Population & Demographics) both claim "urbanisation"; and Theme 1 (AI) and Theme 29 (Sciences) share emerging technology framing.
- Theme 15 (Sustainability & Recycling) should be renamed "Sustainability & Circular Economy" — "recycling" dramatically undersells the scope of circular economy transformation.

SOURCE QUALITY OBSERVATIONS

- The platform over-relies on McKinsey (present in 15+ themes), WEF (present in all themes), and Deloitte — all commercial entities with inherent stakeholder biases. These remain useful but should be balanced with independent academic institutions.
- Critical gaps in the existing lists include: World Bank (financial inclusion, development), IPBES (biodiversity — the IPCC-equivalent for nature, absent from the platform), IUCN (conservation authority), ITU (digital development), and UN Population Division (demographics).
- The source "reuters.com" (Theme 3) is misidentified — Reuters is a news agency, not a research institution. It should be replaced by the Reuters Institute for the Study of Journalism at Oxford.
- Dubai Future Foundation appears in three themes (AI, Business, Sciences) as a commercial government entity with limited academic independence. It has been recommended for removal in all three cases.
- Gartner (commercial analyst) was removed from Security & Safety (Theme 21) where policy-oriented sources are more appropriate.

DESCRIPTION QUALITY OBSERVATIONS

- "Sanctions" in Theme 6 (Global Economy) is politically framed and should be replaced by the academically established concept "geoeconomics."
- "UBI" in Theme 26 (Work & Income) is a specific contested policy proposal; replaced by "income security models" for appropriate analytical neutrality.
- "Big data" in Theme 2 is outdated terminology (peak 2013–2016); updated to "datafication" and "data governance."
- "Smart cities" in Theme 10 is an increasingly contested concept; the revised description centres climate-resilient urban design and urban governance innovation.

PART I: PLATFORM ARCHITECTURE ANALYSIS

1.1 Thematic Coverage Assessment

The 30-theme framework achieves broad coverage of the major futures domains. The six-category structure provides logical organisational clarity. The inclusion of "Values & Ethics" and "Leisure & Lifestyles" as distinct themes reflects a commitment to holistic futures analysis beyond economic and technological determinism — a methodologically sound decision aligned with integral futures studies approaches.

However, the category "Science & Demographics" (Themes 29–30) is an unusual pairing — demographics is a social science domain while Sciences & Research is about the knowledge production system itself. These could arguably occupy separate categories. The pairing may reflect space constraints rather than logical kinship.

1.2 Identified Thematic Overlaps

Theme A	Theme B	Overlap / Recommended Resolution
2: Data & Digitalisation	7: Digital & Sharing Economies	Both reference "platforms." Resolution: Theme 2 owns data governance/sovereignty; Theme 7 owns platform labour and power dynamics. Revised descriptions reflect this.
10: Construction & Urbanisation	30: Population & Demographics	"Urbanisation trends" in Theme 30 description overlaps with Theme 10. Resolution: Remove urbanisation from Theme 30 description; replace with "generational transitions."
1: Artificial Intelligence	29: Sciences & Research	"Quantum computing" in Theme 29 description and AI permeates all scientific research. Resolution: Theme 29 focuses on the scientific knowledge system; AI as a horizontal driver.
14: Climate Change	15: Sustainability	Adjacent but sufficiently distinct: Climate is the physical system; Sustainability is the economic and corporate response. Maintained as separate themes.
24: Health & Wellbeing	29: Sciences & Research	"Biotech research" in Theme 29 overlaps with "biotech" in Theme 24. Resolution: Theme 24 owns clinical/health applications; Theme 29 owns fundamental biotech research frontiers.

1.3 Source Quality Framework

The following quality tiers were applied in evaluating and adding sources:

TIER 1 — Highest Authority:

Intergovernmental bodies and UN agencies (IPCC, WHO, IEA, IRENA, IMF, World Bank, OECD, UN agencies); peer-reviewed flagship journals (Nature, Science, Lancet); national academies (Royal Society, AAAS).

TIER 2 — High Authority:

Leading independent think tanks (Chatham House, RAND, Brookings, Carnegie, CSIS, IISS, SIPRI); major academic institutions (Oxford, Cambridge, MIT, Harvard, Stanford, LSE); European institutions (JRC, EEA, ECFR, Bruegel).

TIER 3 — Solid Specialist:

Specialist research bodies (ILO, FAO, UNEP, ITU, WTO, UNCTAD); futures organisations (IFTF, CIFS, SOIF, Hawaii Research Center); well-established NGOs with research credibility (Ellen MacArthur Foundation, WRI, IUCN, Fairwork).

TIER 4 — Supplementary (use selectively):

Commercial analysts (McKinsey, Deloitte, Gartner, Forrester, BCG); government-funded bodies (Dubai Future Foundation); professional associations (WEF). These sources are retained where specifically authoritative but are not the primary bases for analysis.

PART II: PER-THEME ANALYSIS

Category: Technology & Digital

Theme 1: Artificial Intelligence

FOCUS AREA EVALUATION

Current	*AI, machine learning, autonomous systems*
Revised	**Generative AI, machine learning, autonomous systems, AI governance and regulation, AI safety, human-AI collaboration, societal and economic AI impacts**

Argumentation:

The current description is too narrow and anchored in technical sub-disciplines. It omits the dominant contemporary discourse: generative AI, AI governance and regulation (EU AI Act, NIST AI RMF), AI safety and existential risk, and the critical sociotechnical impacts on labour, democracy, and knowledge production. "Autonomous systems" partially overlaps with "machine learning" and the description overall reflects a 2018–2020 framing of AI rather than the 2024–2026 frontier.

SOURCES REMOVED / REPLACED

Dubai Future Foundation (dubaifuture.gov.ae)	Reason: Government-funded marketing body with limited peer-review; lacks methodological rigor; marginal value given stronger alternatives available.

SOURCES ADDED

Stanford HAI (hai.stanford.edu)	Justification: World-leading human-centred AI research centre; produces authoritative empirical reports on AI adoption, regulation, and societal impact (AI Index).
MIT Technology Review (technologyreview.com)	Justification: Premier technology journalism with rigorous editorial standards; best-in-class coverage of AI developments and governance implications.
Alan Turing Institute (turing.ac.uk)	Justification: UK national AI and data science institute; high-quality policy-relevant research across AI safety, ethics, and applications.
Future of Life Institute (futureoflife.org)	Justification: Specialist focus on AI existential risk and governance; publishes peer-reviewed risk analyses; produced the influential AI moratorium letter.
AI Now Institute (ainowinstitute.org)	Justification: Social science perspective on AI impacts; authoritative on algorithmic discrimination, labour, and surveillance.
OECD (oecd.org)	Justification: AI Principles framework and AI Policy Observatory; authoritative comparative policy data across member states.
Brookings Institution (brookings.edu)	Justification: Rigorous, bipartisan AI governance analysis; significant AI policy influence in US and globally.

FINAL SOURCE LIST (14 sources)

1	Oxford Martin School (oxfordmartin.ox.ac.uk) — AI governance, future risks
2	Stanford HAI (hai.stanford.edu) — human-centred AI, AI Index
3	MIT Technology Review (technologyreview.com) — AI trends and critique
4	Alan Turing Institute (turing.ac.uk) — AI safety and applications
5	CSER Cambridge (cser.ac.uk) — AI existential risk
6	Future of Life Institute (futureoflife.org) — AI governance, risk
7	AI Now Institute (ainowinstitute.org) — social impacts of AI
8	RAND Corporation (rand.org) — AI policy analysis
9	OECD (oecd.org) — AI Principles, policy observatory
10	European Commission (ec.europa.eu) — EU AI Act, regulation
11	Fraunhofer ISI (isi.fraunhofer.de) — AI technology research
12	Brookings Institution (brookings.edu) — AI governance
13	WEF (weforum.org) — AI global impact, adoption
14	IFTF (iftf.org) — long-range AI futures

Theme 2: Data & Digitalisation

FOCUS AREA EVALUATION

Current	*Big data, digital transformation, platforms*
Revised	**Data governance and sovereignty, digital transformation, datafication, open data ecosystems, digital divide and inclusion, privacy and data rights**

Argumentation:

"Big data" is an outdated framing (peak usage circa 2013–2016); it is now subsumed under broader data governance and datafication discourses. "Platforms" creates direct overlap with Theme 7 (Digital & Sharing Economies). The description omits critical contemporary issues: data sovereignty, the digital divide, privacy (GDPR and equivalents), and open data ecosystems. A futures-oriented revision should centre governance and equity alongside transformation.

SOURCES ADDED

ITU (itu.int)	Justification: UN specialised agency for digital and ICT; authoritative on global connectivity, digital development indicators, and spectrum governance.

World Bank (worldbank.org)	Justification: Produces rigorous cross-country analysis of digital dividends, infrastructure, and the economic impacts of digitalisation in developing economies.

Open Data Institute (theodi.org)	Justification: Pioneer institution on open data standards, data trusts, and data governance; directly relevant to the revised description.

Alan Turing Institute (turing.ac.uk)	Justification: Data science and AI policy research; authoritative on data infrastructure and governance.

Bennett Institute for Public Policy, Cambridge (bennettinstitute.cam.ac.uk)	Justification: High-quality public policy research on digital economy, productivity, and data governance.

FINAL SOURCE LIST (12 sources)

1	OECD (oecd.org) — digital economy, data policy
2	European Commission (ec.europa.eu) — EU data strategy, digital markets
3	WEF (weforum.org) — digital transformation
4	McKinsey Global Institute (mckinsey.com) — digitalisation impact
5	ITU (itu.int) — digital development, global connectivity
6	World Bank (worldbank.org) — digital dividends
7	Fraunhofer ISI (isi.fraunhofer.de) — digitalisation research
8	Alan Turing Institute (turing.ac.uk) — data science
9	Open Data Institute (theodi.org) — data governance
10	Bennett Institute, Cambridge (bennettinstitute.cam.ac.uk) — digital public policy
11	Sitra (sitra.fi) — fair data economy
12	Brookings Institution (brookings.edu) — digital economy

Theme 3: Communication & Media

FOCUS AREA EVALUATION

Current	*Telecom, journalism, social media, content*
Revised	**Media transformation, journalism futures, platform governance, disinformation and synthetic media, social media dynamics, digital content ecosystems, trust in information**

Argumentation:

"Telecom" is an industrial/engineering term out of place alongside journalism and social media — it belongs more in Industry & Infrastructure. "Content" is too vague to constitute a futures focus. The description omits the most critical current dimensions: disinformation and synthetic media (deepfakes), platform governance, trust in media, and the collapse of traditional media business models. Critically, the source "reuters.com" is Reuters the news agency — not an analytical institution. The correct source should be the Reuters Institute for the Study of Journalism at Oxford.

SOURCES REMOVED / REPLACED

Reuters (reuters.com)	Reason: This is a news agency, not a research institution. The correct comparable source is the Reuters Institute for the Study of Journalism (reutersinstitute.politics.ox.ac.uk), which should replace it.

SOURCES ADDED

Reuters Institute for Journalism (reutersinstitute.politics.ox.ac.uk)	Justification: THE authoritative institution for journalism research; publishes the annual Digital News Report covering 46 countries; part of Oxford University.

Oxford Internet Institute (oii.ox.ac.uk)	Justification: Leading academic centre studying the intersection of internet, society, and platform media; rigorous empirical research on disinformation, algorithms, and social media.

Poynter Institute (poynter.org)	Justification: Premier journalism ethics and standards institution; authoritative on fact-checking, media trust, and newsroom transformation.

First Draft (firstdraftnews.org)	Justification: Specialist organisation on misinformation and disinformation; produces field-defining research on information disorders.

Knight Foundation (knightfoundation.org)	Justification: Major funder and researcher of journalism innovation and media trust; publishes significant research on news deserts and local media.

FINAL SOURCE LIST (12 sources)

1	Reuters Institute for Journalism (reutersinstitute.politics.ox.ac.uk) — journalism futures, Digital News Report
2	Harvard Nieman Lab (niemanlab.org) — journalism innovation
3	Oxford Internet Institute (oii.ox.ac.uk) — platform media, disinformation
4	Pew Research Center (pewresearch.org) — media consumption, news habits
5	Knight Foundation (knightfoundation.org) — journalism innovation, media trust
6	Poynter Institute (poynter.org) — journalism ethics, fact-checking
7	First Draft (firstdraftnews.org) — disinformation research
8	WEF (weforum.org) — media transformation
9	IFTF (iftf.org) — communication futures
10	CIFS (cifs.dk) — media and communication trends
11	MIT Media Lab (media.mit.edu) — digital media innovation
12	Freedom House (freedomhouse.org) — press freedom

Category: Economy & Business

Theme 4: Business

FOCUS AREA EVALUATION

Current	*Business models, strategy, innovation management*

Revised	**Business model transformation, corporate strategy, entrepreneurship and innovation ecosystems, organizational resilience, stakeholder capitalism, platform disruption**

Argumentation:

Extremely broad and generic — the description could apply to almost any commercial theme in the platform. "Innovation management" is more of a management discipline than a futures focus area. Missing: entrepreneurship ecosystems, corporate longevity and resilience, platform disruption of traditional industry, the stakeholder capitalism debate, and the transformation of value creation models. This theme needs a clearer futures orientation distinguishing it from Theme 8 (Services) and Theme 7 (Digital & Sharing Economies).

SOURCES REMOVED / REPLACED

Dubai Future Foundation (dubaifuture.gov.ae)	Reason: Government promotional body without sufficient academic independence; marginal analytical value for global business futures analysis.

SOURCES ADDED

MIT Sloan Management Review (sloanreview.mit.edu)	Justification: Peer-reviewed management research with strong futures orientation; publishes leading research on business model innovation and digital transformation.
NESTA (nesta.org.uk)	Justification: UK innovation foundation with rigorous methodological standards; produces influential research on innovation ecosystems and business futures.
Kauffman Foundation (kauffman.org)	Justification: THE authoritative institution on entrepreneurship research; provides rigorous data on startup ecosystems and venture dynamics.
Stanford Social Innovation Review (ssir.org)	Justification: Covers the increasingly important space of social enterprise, impact investment, and purpose-driven business — essential for contemporary business futures.
OECD (oecd.org)	Justification: Comparative cross-country data on business dynamism, R&D investment, and innovation policy — essential empirical grounding.

FINAL SOURCE LIST (12 sources)

1	McKinsey Global Institute (mckinsey.com) — business transformation
2	HBR (hbr.org) — strategy and management innovation
3	BCG Henderson Institute (bcg.com) — business strategy
4	MIT Sloan Management Review (sloanreview.mit.edu) — management research
5	WEF (weforum.org) — global business futures
6	OECD (oecd.org) — business dynamism, innovation policy
7	NESTA (nesta.org.uk) — innovation ecosystems
8	Deloitte Insights (deloitte.com) — business transformation

9	Kauffman Foundation (kauffman.org) — entrepreneurship
10	WBCSD (wbcsd.org) — sustainable business models
11	Stanford Social Innovation Review (ssir.org) — purpose-driven business
12	Brookings Institution (brookings.edu) — business economics

Theme 5: Finance & Ownership

FOCUS AREA EVALUATION

Current	*Banking, investment, fintech, ownership models*
Revised	**Banking transformation, investment futures, fintech and CBDCs, financial inclusion, wealth distribution and inequality, alternative and commons-based ownership models**

Argumentation:

The description covers the core well but misses several critical futures dimensions: central bank digital currencies (CBDCs) — now being tested by 130+ countries; decentralised finance (DeFi); the growing wealth inequality discourse; financial inclusion in the Global South; and the transformation of institutional investment toward ESG. "Ownership models" is an interesting and underexplored futures angle (data ownership, commons, cooperatives) that deserves sharper articulation.

SOURCES ADDED

World Bank (worldbank.org)	Justification: Essential for financial inclusion data (Global Findex database), development finance, and sovereign debt analysis in emerging markets.
European Central Bank (ecb.europa.eu)	Justification: Authoritative on CBDC development (digital euro), monetary policy futures, and European financial stability.
Institute for New Economic Thinking (ineteconomics.org)	Justification: Challenges orthodox economic assumptions; produces rigorous research on alternative financial architectures and post-growth economics.
New Economics Foundation (neweconomics.org)	Justification: Specialist in alternative ownership models, commons economics, and financial system reform — directly aligned with the revised description.
Brookings Institution (brookings.edu)	Justification: Rigorous financial regulation analysis and policy research; significant influence on financial system futures.

FINAL SOURCE LIST (13 sources)

1	BIS (bis.org) — central banking, fintech, monetary policy
2	IMF (imf.org) — global financial system stability
3	FSB (fsb.org) — financial stability regulation
4	World Bank (worldbank.org) — financial inclusion

5	OECD (oecd.org) — financial markets and capital
6	European Central Bank (ecb.europa.eu) — CBDC, monetary innovation
7	McKinsey (mckinsey.com) — financial services transformation
8	WEF (weforum.org) — financial system futures
9	Chatham House (chathamhouse.org) — global finance and geopolitics
10	Institute for New Economic Thinking (ineteconomics.org) — alternative economic architectures
11	New Economics Foundation (neweconomics.org) — alternative ownership
12	Sitra (sitra.fi) — fair economy, ownership models
13	Brookings Institution (brookings.edu) — financial regulation

Theme 6: Global Economy

FOCUS AREA EVALUATION

Current	*Trade, macroeconomics, supply chains, sanctions*
Revised	**Global trade, macroeconomics, supply chain transformation, geoeconomics, deglobalisation, emerging economies, sovereign debt, global inequality**

Argumentation:

"Sanctions" is a highly specific, politically charged policy instrument — listing it as a defining focus of global economic futures is reductive and potentially biased toward a Western policy lens. It should be replaced by "geoeconomics," which is the broader, academically established concept. The description also misses critical futures themes: deglobalisation and reshoring, the rise of emerging market blocs (BRICS expansion), sovereign debt crises in the Global South, and the role of climate in disrupting economic systems.

SOURCES ADDED

World Bank (worldbank.org)	Justification: Essential for development economics, global poverty data, and cross-country economic analysis; conspicuously absent from the current list.
WTO (wto.org)	Justification: THE authority on international trade rules and data; indispensable for global economy futures.
UNCTAD (unctad.org)	Justification: UN body specialising in trade and development; critical perspective from the Global South on global economic governance.
Peterson Institute for International Economics (piie.com)	Justification: Leading independent research institution on international trade, currency, and macroeconomics; highly respected for rigorous econometric analysis.
Bruegel (bruegel.org)	Justification: Premier European economic think tank; essential for European geoeconomic perspective and monetary analysis.

Atlantic Council (atlanticcouncil.org)	Justification: Strong coverage of economic statecraft, trade security, and geoeconomics; complementary to Chatham House perspective.

FINAL SOURCE LIST (13 sources)

1	IMF (imf.org) — World Economic Outlook
2	OECD (oecd.org) — economic analysis and indicators
3	World Bank (worldbank.org) — development economics
4	WTO (wto.org) — global trade
5	UNCTAD (unctad.org) — trade and development, Global South
6	WEF (weforum.org) — global competitiveness
7	Chatham House (chathamhouse.org) — geoeconomics
8	Peterson Institute (piie.com) — international economics
9	Brookings Institution (brookings.edu) — global economy
10	Atlantic Council (atlanticcouncil.org) — economic statecraft
11	Bruegel (bruegel.org) — European economic policy
12	NIC (dni.gov) — global economic trends (Global Trends report)
13	Millennium Project (millennium-project.org) — global economic challenges

Theme 7: Digital & Sharing Economies

FOCUS AREA EVALUATION

Current	*Platform economy, gig work, collaborative models*
Revised	**Platform economy, algorithmic management and gig labour rights, peer-to-peer models, platform monopolies, collaborative and commons-based digital economies, digital labour governance**

Argumentation:

There is a significant structural overlap with Theme 2 (Data & Digitalisation) — "platform economy" and "platforms" appear in both. The distinctive identity of this theme should be the labour and power dynamics of platform capitalism: algorithmic management, gig worker rights, platform monopolisation, and the governance of digital labour. "Collaborative models" is vague; it should be sharpened to peer-to-peer economies and digital commons.

SOURCES ADDED

ILO (ilo.org)	Justification: THE authority on labour standards; essential for gig work governance, platform labour rights, and the future of digital employment globally.

Fairwork Foundation (fair.work)	Justification: Specialist research on platform worker conditions across 40+ countries; produces annual ratings of platform labour standards — uniquely relevant.

Oxford Internet Institute (oii.ox.ac.uk)	Justification: Rigorous academic research on platform economy dynamics, digital labour, and gig work; produces the iLabour Project data.

Brookings Institution (brookings.edu)	Justification: Authoritative analysis of digital competition, platform regulation, and antitrust in the digital economy.

FINAL SOURCE LIST (12 sources)

1	OECD (oecd.org) — platform economy, digital markets
2	European Commission (ec.europa.eu) — Digital Markets Act, platform regulation
3	ILO (ilo.org) — gig work, digital labour standards
4	McKinsey Global Institute (mckinsey.com) — platform economy
5	WEF (weforum.org) — sharing economy futures
6	Oxford Internet Institute (oii.ox.ac.uk) — platform dynamics, digital labour
7	IFTF (iftf.org) — future platforms and work
8	Fairwork Foundation (fair.work) — platform labour rights
9	Sitra (sitra.fi) — fair data economy
10	Deloitte (deloitte.com) — platform business models
11	Brookings Institution (brookings.edu) — digital competition
12	MIT Media Lab (media.mit.edu) — collaborative digital models

Theme 8: Services

FOCUS AREA EVALUATION

Current	*Service design, customer experience, servitization*
Revised	**Service economy transformation, AI-augmented services, public service innovation, service design futures, platform-mediated services, knowledge-intensive services**

Argumentation:

"Servitization" is a narrow manufacturing concept (adding services to physical products) — it is a sub-discipline, not a defining focus for an entire futures theme. "Customer experience" is an operational, commercial framing rather than a futures lens. This theme suffers from identity diffusion: it sits between Business (4), Industry (9), and Digital Economy (7) without a sharp futures-oriented angle. It should be reframed around the transformation of the service economy as a whole — including AI-driven services, public service innovation, and platform-mediated services.

SOURCES ADDED

NESTA (nesta.org.uk)	Justification: Leading source on public service innovation and the future of welfare state services; particularly strong on health, education, and local government innovation.

GovLab, NYU (thegovlab.org)	Justification: Specialist on open and data-driven governance and public service redesign; highly relevant for the public services dimension of this theme.

RSA (thersa.org)	Justification: Royal Society of Arts; produces futures-oriented research on service innovation, social purpose, and the economy of the future.

Fraunhofer ISI (isi.fraunhofer.de)	Justification: Strong applied research on service innovation and the service economy in industrial transformation contexts.

FINAL SOURCE LIST (12 sources)

1	McKinsey Global Institute (mckinsey.com) — service sector transformation
2	OECD (oecd.org) — services trade and innovation
3	WEF (weforum.org) — services economy futures
4	HBR (hbr.org) — service innovation management
5	Deloitte (deloitte.com) — service transformation
6	Forrester (forrester.com) — customer experience and service futures
7	NESTA (nesta.org.uk) — public service innovation
8	GovLab, NYU (thegovlab.org) — data-driven public services
9	RSA (thersa.org) — service futures
10	Fraunhofer ISI (isi.fraunhofer.de) — service research
11	Sitra (sitra.fi) — service innovation
12	PwC Strategy& (pwc.com) — service industry transformation

Category: Industry & Infrastructure

Theme 9: Industry & Manufacturing

FOCUS AREA EVALUATION

Current	*Smart factories, automation, industrial IoT*
Revised	**Smart manufacturing and Industry 5.0, industrial automation, industrial decarbonisation, reshoring and industrial resilience, additive manufacturing, green industrial policy**

Argumentation:

The description is anchored in Industry 4.0 technology framing (2015–2020 era). It misses the now-critical dimension of industrial decarbonisation (net-zero manufacturing), the emerging Industry 5.0 concept (human-centric, resilient, sustainable), reshoring and reindustrialisation trends driven by geopolitical fragmentation, and additive manufacturing. The description should be updated to reflect the post-2022 industrial transformation agenda.

SOURCES ADDED

European Commission — Industrial Strategy (ec.europa.eu)	Justification: The EU Green Deal Industrial Plan and the Critical Raw Materials Act represent defining frameworks for European industrial futures; indispensable.

UNIDO (unido.org)	Justification: UN Industrial Development Organization; essential for global industrial development, sustainable industrialisation, and manufacturing in emerging economies.

IEA (iea.org)	Justification: Increasingly authoritative on industrial decarbonisation — clean hydrogen, electrification of heat, and net-zero industrial pathways.

Brookings Institution (brookings.edu)	Justification: Strong analysis of reindustrialisation, industrial policy revival, and manufacturing futures in advanced economies.

BCG (bcg.com)	Justification: Henderson Institute publishes rigorous analysis on Industry 4.0/5.0 transition, automation economics, and green manufacturing.

FINAL SOURCE LIST (12 sources)

1	WEF (weforum.org) — manufacturing futures, Industry 4.0/5.0
2	Fraunhofer ISI (isi.fraunhofer.de) — industrial technology research
3	OECD (oecd.org) — industrial policy
4	European Commission (ec.europa.eu) — EU Industrial Strategy
5	UNIDO (unido.org) — sustainable industrial development
6	McKinsey Global Institute (mckinsey.com) — manufacturing transformation
7	IEA (iea.org) — industrial decarbonisation
8	Deloitte (deloitte.com) — manufacturing trends
9	BCG (bcg.com) — Industry 4.0/5.0, green manufacturing
10	Brookings Institution (brookings.edu) — reindustrialisation
11	NESTA (nesta.org.uk) — industrial innovation
12	MIT Industrial Performance Center (ipc.mit.edu) — manufacturing competitiveness

Theme 10: Construction & Urbanisation

FOCUS AREA EVALUATION

Current	*Smart cities, urban planning, building tech*
Revised	**Urban futures and climate-resilient cities, housing systems, smart urban infrastructure, urban planning innovation, construction technology, built environment sustainability**

Argumentation:

"Smart cities" is an increasingly contested concept that has been substantially critiqued for surveillance risks, vendor lock-in, and techno-solutionism (see Sidewalk Toronto failure). A more sophisticated futures framing should include: climate-resilient urban design, housing affordability crises, the 15-minute city concept, informal settlements, and urban governance innovation. "Building tech" is too narrow and industry-specific. There is also thematic overlap with Theme 30 (Population & Demographics), which mentions "urbanisation trends" — this theme should own "urban futures" clearly.

SOURCES ADDED

UN Habitat (unhabitat.org)	Justification: THE UN agency for sustainable urban development; produces World Cities Reports; essential for global urbanisation data and housing policy.
C40 Cities (c40.org)	Justification: Network of 100 cities committed to climate action; authoritative on urban climate adaptation, net-zero cities, and urban resilience.
LSE Cities (lse.ac.uk/cities)	Justification: Academic urban research centre producing rigorous analysis on urban governance, inequality, and planning futures.
WRI Ross Centre for Sustainable Cities (wri.org)	Justification: Research on urban mobility, housing, and climate-resilient urban infrastructure; strong Global South perspective.
ICLEI (iclei.org)	Justification: Global network of 2,500+ local governments for sustainability; authoritative on local climate action and urban resilience policies.

FINAL SOURCE LIST (13 sources)

1	UN Habitat (unhabitat.org) — urban development, housing
2	OECD (oecd.org) — urbanisation policy
3	C40 Cities (c40.org) — climate-resilient cities
4	WEF (weforum.org) — urban transformation
5	LSE Cities (lse.ac.uk/cities) — urban research
6	WRI Ross Centre (wri.org) — sustainable cities
7	European Commission (ec.europa.eu) — EU urban agenda
8	McKinsey Global Institute (mckinsey.com) — cities futures
9	Arup (arup.com) — built environment futures
10	Oxford Martin School (oxfordmartin.ox.ac.uk) — urbanisation
11	ICLEI (iclei.org) — local sustainability
12	Urban Land Institute (uli.org) — real estate and cities
13	Brookings Institution (brookings.edu) — metropolitan futures

Theme 11: Energy

FOCUS AREA EVALUATION

Current	*Renewables, grids, energy storage, nuclear*
Revised	**Renewable energy transition, energy grids and storage, nuclear renaissance, hydrogen economy, energy security, just transition, energy geopolitics**

Argumentation:

Technically sound but misses the strategic dimensions that now dominate energy futures discourse: energy security (dramatically elevated post-2022), the hydrogen economy, energy poverty and just transition, and energy geopolitics. The description is primarily supply-side and technological; it should acknowledge the demand side and governance dimension. Also, "Club of Rome" is a marginal addition to this theme given the presence of IEA, IRENA, and Tyndall Centre.

SOURCES REMOVED / REPLACED

Club of Rome (clubofrome.org)	Reason: Foundational but primarily systemic/philosophical; for the Energy theme specifically, technical and policy sources (IEA, IRENA, Tyndall) provide superior depth. Club of Rome is better positioned in Sustainability or Climate themes.

SOURCES ADDED

Rocky Mountain Institute (rmi.org)	Justification: World-leading clean energy transition research; authoritative on electrification, clean hydrogen, and fossil fuel phase-out economics.
Oxford Institute for Energy Studies (oxfordenergy.org)	Justification: Premier academic research on energy markets, geopolitics of energy, gas markets, and transition economics.
Chatham House (chathamhouse.org)	Justification: Essential for energy security and geopolitics dimension, especially post-Ukraine war energy system transformation.
Ember (ember-climate.org)	Justification: Data-driven, transparent electricity system monitoring; publishes authoritative annual Global Electricity Review with open data.
Columbia SIPA Center on Global Energy Policy (energypolicy.columbia.edu)	Justification: Leading academic policy centre on energy geopolitics, transition finance, and energy security.

FINAL SOURCE LIST (12 sources)

1	IEA (iea.org) — global energy outlook
2	IRENA (irena.org) — renewable energy
3	WEF (weforum.org) — energy transition
4	Rocky Mountain Institute (rmi.org) — clean energy transition
5	Oxford Institute for Energy Studies (oxfordenergy.org) — energy markets, geopolitics
6	Fraunhofer ISI (isi.fraunhofer.de) — energy technology research
7	Tyndall Centre (tyndall.ac.uk) — energy decarbonisation, carbon budgets

8	BloombergNEF (bloombergnef.com) — energy market data
9	Chatham House (chathamhouse.org) — energy security
10	WRI (wri.org) — energy transition
11	Ember (ember-climate.org) — electricity data analytics
12	Columbia Global Energy Policy (energypolicy.columbia.edu) — energy geopolitics

Theme 12: Freight & Logistics

FOCUS AREA EVALUATION

Current	*Supply chains, last-mile, autonomous freight*
Revised	**Supply chain resilience and transformation, freight decarbonisation, autonomous and digital logistics, nearshoring, maritime and port futures, last-mile innovation**

Argumentation:

"Last-mile" is a specific logistics industry operations term, not a futures-defining concept. "Autonomous freight" is an emerging technology, not a framing principle. The description omits: supply chain resilience (dramatically elevated post-COVID), nearshoring and geopolitical supply chain restructuring, maritime shipping (80% of global trade), digital supply chain platforms, and the critical decarbonisation imperative for freight.

SOURCES ADDED

IMO (imo.org)	Justification: International Maritime Organization; authoritative on global shipping regulation, decarbonisation of maritime transport, and port futures.
UNCTAD (unctad.org)	Justification: Publishes the annual Review of Maritime Transport; essential data on global shipping and trade logistics.
International Transport Forum (itf-oecd.org)	Justification: OECD-affiliated body; authoritative on transport policy, freight efficiency, and decarbonisation pathways.
MIT Centre for Transportation and Logistics (ctl.mit.edu)	Justification: Leading academic research on supply chain resilience, autonomous logistics, and freight innovation.

FINAL SOURCE LIST (12 sources)

1	WEF (weforum.org) — supply chain futures
2	OECD/ITF (itf-oecd.org) — transport and logistics policy
3	IEA (iea.org) — freight decarbonisation
4	IMO (imo.org) — maritime shipping regulation
5	UNCTAD (unctad.org) — shipping and trade data
6	World Bank (worldbank.org) — trade logistics

7	McKinsey (mckinsey.com) — supply chain transformation
8	MIT Centre for Transportation and Logistics (ctl.mit.edu) — supply chain research
9	Fraunhofer ISI (isi.fraunhofer.de) — logistics research
10	ILO (ilo.org) — supply chain labour
11	Deloitte (deloitte.com) — supply chain strategy
12	Gartner (gartner.com) — supply chain technology trends

Theme 13: Private & Public Transport

FOCUS AREA EVALUATION

Current	*Mobility-as-a-Service, EVs, public transit*
Revised	**Electric vehicles and decarbonisation, autonomous mobility, Mobility-as-a-Service, shared and active mobility, public transit futures, aviation decarbonisation, mobility equity**

Argumentation:

Reasonably focused but needs enrichment. "MaaS" as a concept has partially underdelivered on early promises — the description should be broadened to include autonomous vehicles (now entering commercial deployment), active mobility (cycling, walking — policy-relevant post-COVID), and critically, aviation decarbonisation (one of the hardest-to-abate sectors). The equity dimension of mobility access is also absent.

SOURCES ADDED

ICCT (theicct.org)	Justification: International Council on Clean Transportation; THE authority on vehicle emissions standards, EV policy, and clean aviation — essential for this theme.
International Transport Forum (itf-oecd.org)	Justification: OECD transport policy body; authoritative comparative data on mobility systems and future transport scenarios.
Transport & Environment (transport-environment.org)	Justification: Leading European NGO on transport decarbonisation; rigorous analysis of EV policy, aviation, and shipping.
UITP (uitp.org)	Justification: International Association of Public Transport; THE authority on public transit futures, data, and urban mobility.
Rocky Mountain Institute (rmi.org)	Justification: Clean mobility transition research, EV economics, and clean aviation; complements ICCT and T&E perspectives.

FINAL SOURCE LIST (12 sources)

1	IEA (iea.org) — EV outlook, transport energy
2	ICCT (theicct.org) — clean transportation standards
3	International Transport Forum (itf-oecd.org) — transport policy

4	Transport & Environment (transport-environment.org) — EU transport decarbonisation
5	WEF (weforum.org) — mobility futures
6	McKinsey (mckinsey.com) — mobility transformation
7	OECD (oecd.org) — transport policy
8	UITP (uitp.org) — public transit
9	Rocky Mountain Institute (rmi.org) — clean mobility
10	Deloitte (deloitte.com) — mobility trends
11	RAND Corporation (rand.org) — transport scenarios
12	IFTF (iftf.org) — mobility futures

Category: Environment & Resources

Theme 14: Climate Change

FOCUS AREA EVALUATION

Current	*Mitigation, adaptation, carbon markets*
Revised	**Climate mitigation and adaptation, carbon markets and removal technologies, climate tipping points, climate finance and loss and damage, climate justice, geoengineering governance**

Argumentation:

Good core framing but could be sharpened. "Carbon markets" is one instrument; the description should include carbon dioxide removal (CDR), climate tipping points (increasingly central to scientific discourse), climate finance and loss and damage (now a formal UNFCCC agenda item), and the justice dimensions of climate action. The current 14-source list is already within range — pruning is needed for quality consistency. Tellus Institute and Millennium Project are weaker here than the specialist climate institutions.

SOURCES REMOVED / REPLACED

Tellus Institute (tellus.org)	Reason: Less specialist and less authoritative than the other climate institutions in this list; IPCC, PIK, and Stockholm Resilience Centre provide superior scientific depth.

Millennium Project (millennium-project.org)	Reason: General foresight body without specialist climate science expertise; less authoritative than IPCC, WMO, and PIK for this specific theme.

SOURCES ADDED

UNEP (unep.org)	Justification: UN Environment Programme; publishes the Emissions Gap Report and Adaptation Gap Report — foundational annual assessments of climate action progress.

UNFCCC (unfccc.int)	Justification: The climate policy framework institution; essential for tracking international commitments, NDCs, and loss and damage negotiations.

FINAL SOURCE LIST (14 sources)

1	IPCC (ipcc.ch) — climate science authority
2	UNEP (unep.org) — Emissions Gap and Adaptation Gap Reports
3	WMO (wmo.int) — weather and climate data
4	PIK Potsdam (pik-potsdam.de) — climate tipping points, earth system
5	Stockholm Resilience Centre (stockholmresilience.org) — planetary boundaries
6	Tyndall Centre (tyndall.ac.uk) — carbon budgets, decarbonisation
7	Oxford Martin School (oxfordmartin.ox.ac.uk) — climate policy
8	WEF (weforum.org) — Global Risks Report, climate action
9	IEA (iea.org) — energy-climate nexus
10	Future Earth (futureearth.org) — climate systems science
11	NASA Climate (climate.nasa.gov) — climate data and monitoring
12	UNFCCC (unfccc.int) — climate policy framework
13	WRI (wri.org) — climate solutions
14	Sitra (sitra.fi) — climate megatrends

Theme 15: Sustainability & Recycling

FOCUS AREA EVALUATION

Current	*Circular economy, waste, ESG reporting*
Revised	**Circular economy, sustainable production and consumption, regenerative systems, ESG and corporate sustainability transformation, material flows, biodegradable innovation**

Argumentation:

The theme name itself is misleading: "Recycling" is a narrow end-of-pipe intervention that significantly undersells the theme scope. The theme is fundamentally about circular economy and corporate sustainability transformation. "ESG reporting" is currently one of the most contested dimensions of sustainability — greenwashing litigation, the SEC rule, and European CSRD create important futures tensions that require balanced treatment. Recommend renaming the theme to "Sustainability & Circular Economy" and broadening the description to include regenerative approaches and sustainable consumption.

SOURCES ADDED

UNEP (unep.org)	Justification: UN Environment Programme; authority on global sustainability frameworks and the Global Resources Outlook; conspicuously absent.

GRI (globalreporting.org)	Justification: Global Reporting Initiative; THE standard-setter for ESG and sustainability reporting — essential given "ESG reporting" is in the description.

CDP (cdp.net)	Justification: The world's largest corporate environmental disclosure system; produces authoritative data on corporate climate, water, and forests performance.

Science Based Targets initiative (sciencebasedtargets.org)	Justification: Defines corporate net-zero pathways aligned with climate science; increasingly the benchmark for credible corporate sustainability commitments.

FINAL SOURCE LIST (13 sources)

1	Ellen MacArthur Foundation (ellenmacarthurfoundation.org) — circular economy
2	European Commission (ec.europa.eu) — Circular Economy Action Plan
3	UNEP (unep.org) — global sustainability frameworks
4	WEF (weforum.org) — ESG, sustainability transformation
5	OECD (oecd.org) — sustainability policy
6	WBCSD (wbcsd.org) — corporate sustainability
7	GRI (globalreporting.org) — ESG reporting standards
8	Stockholm Resilience Centre (stockholmresilience.org) — sustainability science
9	Club of Rome (clubofrome.org) — systemic sustainability
10	CDP (cdp.net) — corporate environmental data
11	WRI (wri.org) — resource efficiency
12	Science Based Targets (sciencebasedtargets.org) — corporate net-zero
13	Sitra (sitra.fi) — circular economy

Theme 16: Nature

FOCUS AREA EVALUATION

Current	*Biodiversity, ecosystems, land use*
Revised	**Biodiversity loss and recovery, ecosystem services, natural capital valuation, nature-based solutions, rewilding, land and ocean governance, biodiversity finance**

Argumentation:

The description is accurate but surface-level. It omits critical contemporary dimensions: nature-based solutions (NbS) — now a mainstream climate and development concept; natural capital valuation (TNFD framework); rewilding; ocean biodiversity; and the biodiversity-finance nexus (the Kunming-Montreal Global Biodiversity Framework, 2022). "Land use" is covered partially by other themes (Food, Construction). The key gap is the absence of IPBES — the biodiversity equivalent of IPCC — and IUCN from both description and sources.

SOURCES ADDED

IPBES (ipbes.net)	Justification: Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services; THE biodiversity equivalent of IPCC; produces

	authoritative global biodiversity assessments. Absence from the current list is a critical gap.

IUCN (iucn.org)	Justification: International Union for Conservation of Nature; publishes the Red List of Threatened Species and authoritative conservation standards; indispensable for any Nature theme.

TNFD (tnfd.global)	Justification: Taskforce on Nature-related Financial Disclosures; emerging standard-setter for corporate nature risk disclosure; highly futures-relevant.

UNEP-WCMC (unep-wcmc.org)	Justification: World Conservation Monitoring Centre; key institution for biodiversity data, protected area monitoring, and natural capital assessment.

Natural Capital Coalition / Capitals Coalition (capitalscoalition.org)	Justification: THE authority on natural capital valuation frameworks; directly relevant to the biodiversity-finance nexus.

FINAL SOURCE LIST (12 sources)

1	IPBES (ipbes.net) — biodiversity science authority
2	IUCN (iucn.org) — conservation status, nature policy
3	UN CBD (cbd.int) — Kunming-Montreal Biodiversity Framework
4	Stockholm Resilience Centre (stockholmresilience.org) — planetary boundaries
5	WEF (weforum.org) — nature and biodiversity
6	Future Earth (futureearth.org) — biodiversity research
7	UNEP-WCMC (unep-wcmc.org) — nature monitoring
8	WRI (wri.org) — forests and biodiversity
9	FAO (fao.org) — land ecosystems
10	Oxford Martin School (oxfordmartin.ox.ac.uk) — biodiversity
11	TNFD (tnfd.global) — nature-related financial disclosure
12	Capitals Coalition (capitalscoalition.org) — natural capital valuation

Theme 17: Food

FOCUS AREA EVALUATION

Current	*Agri-tech, food security, alternative proteins*
Revised	**Food systems transformation, agricultural innovation, food security and sovereignty, alternative and cellular proteins, agroecology, food waste, nutrition futures**

Argumentation:

"Agri-tech" is supply-side and technological in framing; it misses the political economy of food systems (food sovereignty, agroecology, corporate concentration), food waste (one-third of food produced is wasted), nutrition and non-communicable diseases, and the transformative potential of precision fermentation and cellular agriculture. The description should signal the systemic transformation of food rather than specific technologies.

SOURCES ADDED

World Food Programme (wfp.org)	Justification: UN agency addressing hunger; essential for food security in fragile contexts, climate-conflict-food nexus, and humanitarian food systems.
CGIAR (cgiar.org)	Justification: THE global network of agricultural research centres; produces foundational science on crop improvement, food systems transformation, and climate-resilient agriculture.
IFPRI (ifpri.org)	Justification: International Food Policy Research Institute; authoritative on global food policy, nutrition, agricultural markets, and food security modelling.
EAT Forum (eatforum.org)	Justification: Publishes the landmark EAT-Lancet Commission on sustainable diets; bridges food science, planetary health, and policy.
Good Food Institute (gfi.org)	Justification: THE authority on alternative proteins (plant-based, fermentation, cultivated meat); produces rigorous market and science reports — essential for "alternative proteins" in the description.
IPES-Food (ipes-food.org)	Justification: International Panel of Experts on Sustainable Food Systems; produces independent, critical analysis of food system governance and agroecology.

FINAL SOURCE LIST (12 sources)

1	FAO (fao.org) — food security, agriculture
2	WFP (wfp.org) — food security, hunger
3	CGIAR (cgiar.org) — agricultural research
4	IFPRI (ifpri.org) — food policy
5	WEF (weforum.org) — food systems
6	OECD (oecd.org) — agriculture and food
7	WRI (wri.org) — food systems, land use
8	EAT Forum (eatforum.org) — sustainable diets
9	Good Food Institute (gfi.org) — alternative proteins
10	Oxford Martin School (oxfordmartin.ox.ac.uk) — food futures
11	IPES-Food (ipes-food.org) — food system governance
12	Nature (nature.com) — food systems research

Theme 18: Water & Oceans

FOCUS AREA EVALUATION

Current	*Water scarcity, marine economy, blue tech*
Revised	**Water security and freshwater scarcity, ocean governance and health, marine economy, ocean acidification, water-food-energy nexus, desalination and water technology**

Argumentation:

"Blue tech" is an emerging industry term without established academic definition; it should be replaced with more substantive framings. The description should include ocean governance (UNCLOS, high seas treaty), ocean acidification and its impacts on marine food chains, the water-food-energy nexus, and freshwater ecosystem collapse. "Marine economy" is commercial framing that should be balanced with conservation and governance dimensions.

SOURCES ADDED

UN Water (unwater.org)	Justification: The UN inter-agency coordination body on freshwater; publishes the World Water Development Report — THE global water authority.
World Water Council (worldwatercouncil.org)	Justification: International water policy platform; produces strategic analysis on water governance and scarcity — organises World Water Forum.
Oceana (oceana.org)	Justification: Largest ocean conservation organisation; produces rigorous scientific reports on ocean health, overfishing, and marine protection.
UNESCO-IHP (en.unesco.org/themes/water-security)	Justification: UNESCO International Hydrological Programme; authority on freshwater resources science and governance.
WWF (wwf.org)	Justification: Significant freshwater programme with authoritative Living Planet Index; essential for freshwater ecosystem monitoring.

FINAL SOURCE LIST (13 sources)

1	UN Water (unwater.org) — water governance, WWDR
2	OECD (oecd.org) — water policy
3	WEF (weforum.org) — water and ocean futures
4	UNEP (unep.org) — ocean environment
5	WMO (wmo.int) — hydrological data
6	IIASA (iiasa.ac.at) — water scenarios
7	World Water Council (worldwatercouncil.org) — water policy
8	UNESCO-IHP (en.unesco.org) — water resources

9	Oceana (oceana.org) — ocean conservation
10	WWF (wwf.org) — freshwater ecosystems
11	Nature (nature.com) — ocean and water research
12	Strategic Foresight Group (strategicforesight.com) — water diplomacy
13	Millennium Project (millennium-project.org) — water futures

Category: Society & Governance

Theme 19: Public Governance

FOCUS AREA EVALUATION

Current	*Regulation, democracy, public-sector innovation*
Revised	**Democratic governance futures, public-sector innovation, digital and AI governance, regulatory transformation, open government and citizen participation, institutional trust**

Argumentation:

The description is adequate but lacks the contemporary urgency of digital governance challenges. "Regulation" is broad — the most critical futures dimension is the governance of emerging technologies (AI Act, DSA, DMA). "Democracy" should be expanded to include democratic backsliding and resilience. Missing: open government, algorithmic accountability, citizen participation technologies, regulatory agility, and trust in public institutions — a defining futures challenge.

SOURCES ADDED

V-Dem Institute (v-dem.net)	Justification: Varieties of Democracy project; THE authoritative global democracy data source; essential for evidence-based governance analysis.
GovLab, NYU (thegovlab.org)	Justification: Leading institution on open and data-driven governance; publishes rigorous research on digital governance innovation and participatory democracy.
Open Government Partnership (opengovpartnership.org)	Justification: International initiative for transparent and accountable government; tracks and compares open government commitments across 75+ countries.
Transparency International (transparency.org)	Justification: THE authority on corruption and governance quality globally; Corruption Perceptions Index is a critical benchmarking tool.
Harvard Kennedy School (hks.harvard.edu)	Justification: Leading academic institution for public policy and governance research; produces highly cited work on regulatory design and public management.

FINAL SOURCE LIST (13 sources)

1	OECD (oecd.org) — governance innovation, open government

2	V-Dem Institute (v-dem.net) — democracy data
3	Freedom House (freedomhouse.org) — democracy indices
4	GovLab, NYU (thegovlab.org) — digital governance
5	Open Government Partnership (opengovpartnership.org) — transparency
6	WEF (weforum.org) — governance futures
7	Transparency International (transparency.org) — anti-corruption
8	European Commission (ec.europa.eu) — regulatory innovation
9	Chatham House (chathamhouse.org) — governance
10	RAND Corporation (rand.org) — governance policy
11	Harvard Kennedy School (hks.harvard.edu) — public policy
12	SOIF (soif.org.uk) — governance foresight
13	Sitra (sitra.fi) — public innovation

Theme 20: Geopolitics

FOCUS AREA EVALUATION

Current	*International relations, conflicts, alliances*
Revised	**Great power competition, multipolar world order, economic statecraft and tech nationalism, regional alliances, conflict dynamics, multilateral system transformation, climate geopolitics**

Argumentation:

The description is generic and could apply to a 1990s IR textbook. It needs to reflect the distinctive characteristics of 2024–2040 geopolitical futures: the return of great power competition, multipolar world order formation, economic statecraft (tech nationalism, industrial policy competition), the fragmentation of multilateral institutions, and the intersection of climate and geopolitics. "Conflicts, alliances" is historically framed rather than futures-oriented.

SOURCES ADDED

Carnegie Endowment for International Peace (carnegieendowment.org)	Justification: One of the most respected international relations think tanks; rigorous analysis across great power relations, democracy, and regional security.

IISS (iiss.org)	Justification: International Institute for Strategic Studies; THE authority on defence, military balance, and strategic affairs; publishes the Military Balance annually.

CFR (cfr.org)	Justification: Council on Foreign Relations; highly influential US foreign policy institution; produces rigorous geopolitical analysis and scenario planning.

ECFR (ecfr.eu)	Justification: European Council on Foreign Relations; essential European geopolitical perspective; strong on EU-China, EU-US, and European strategic autonomy.

SIPRI (sipri.org)	Justification: Stockholm International Peace Research Institute; THE authority on military expenditures, arms control, and conflict data — essential empirical grounding.

Atlantic Council (atlanticcouncil.org)	Justification: Strong on transatlantic relations, strategic competition, and economic statecraft; complements Chatham House and ECFR.

CSIS (csis.org)	Justification: Center for Strategic and International Studies; authoritative US strategic affairs think tank; strong on Indo-Pacific and tech competition.

FINAL SOURCE LIST (12 sources)

1	Chatham House (chathamhouse.org) — international relations
2	Carnegie Endowment (carnegieendowment.org) — international peace
3	CFR (cfr.org) — global governance
4	IISS (iiss.org) — strategic and military affairs
5	RAND Corporation (rand.org) — geopolitical analysis
6	ECFR (ecfr.eu) — European geopolitics
7	NIC (dni.gov) — Global Trends intelligence assessment
8	Atlantic Council (atlanticcouncil.org) — transatlantic affairs
9	SIPRI (sipri.org) — peace, security, conflict data
10	CSIS (csis.org) — strategic studies
11	Strategic Foresight Group (strategicforesight.com) — conflict prevention
12	Millennium Project (millennium-project.org) — geopolitical foresight

Theme 21: Security & Safety

FOCUS AREA EVALUATION

Current	*Cybersecurity, defence, resilience, privacy*
Revised	**Cybersecurity, hybrid warfare, critical infrastructure resilience, defence technology, biosecurity, information operations, privacy and surveillance, societal resilience**

Argumentation:

Broadly adequate but misses the most important emerging security dimensions: hybrid warfare (information operations, election interference, critical infrastructure attacks) — which sits between traditional "defence" and "cybersecurity"; biosecurity (highlighted post-COVID as existential risk); space security; and the distinction between personal privacy and national surveillance. "Safety" in

the theme name implies physical safety, which overlaps with transportation and industrial safety themes.

SOURCES REMOVED / REPLACED

Gartner (gartner.com)	Reason: Commercial analyst firm focused on enterprise IT security products; less appropriate for a futures security theme that requires strategic, policy-oriented sources.

SOURCES ADDED

SIPRI (sipri.org)	Justification: Essential for defence expenditure data, arms control, and conflict monitoring; THE peace and security data authority.
IISS (iiss.org)	Justification: Authoritative on military technology, cyber conflict, and strategic security; produces annual Military Balance and Cyber Capabilities reports.
ENISA (enisa.europa.eu)	Justification: EU Agency for Cybersecurity; the regulatory and technical authority on European cyber resilience and critical infrastructure protection.
NATO (nato.int)	Justification: Essential for collective defence futures, cyber defence doctrine, and hybrid warfare policy frameworks.
Hybrid CoE (hybridcoe.fi)	Justification: European Centre of Excellence for Countering Hybrid Threats (Helsinki); THE specialist institution on hybrid threats — particularly relevant given the European security context.
Belfer Center, Harvard (belfercenter.org)	Justification: Leading academic institution on international security; strong on cyber conflict, nuclear security, and biosecurity.

FINAL SOURCE LIST (12 sources)

1	RAND Corporation (rand.org) — security analysis
2	IISS (iiss.org) — strategic and military security
3	SIPRI (sipri.org) — security and conflict data
4	Chatham House (chathamhouse.org) — security policy
5	NIC (dni.gov) — threat assessment
6	ENISA (enisa.europa.eu) — European cybersecurity
7	WEF (weforum.org) — Global Risks, cyber
8	NATO (nato.int) — collective security
9	Hybrid CoE (hybridcoe.fi) — hybrid threats
10	Belfer Center (belfercenter.org) — international security
11	Atlantic Council (atlanticcouncil.org) — cybersecurity, defence

12	CSER Cambridge (cser.ac.uk) — existential security risks

Theme 22: Values & Ethics

FOCUS AREA EVALUATION

Current	*Trust, transparency, corporate responsibility*
Revised	**Digital and AI ethics, bioethics, intergenerational justice, social trust, value pluralism, corporate ethics and accountability, ethics of emerging technologies**

Argumentation:

"Corporate responsibility" frames ethics primarily through a business lens, which is too narrow. "Trust" and "transparency" overlap significantly. The description misses the central ethical challenges of the futures landscape: AI ethics and algorithmic fairness, bioethics (gene editing, synthetic biology), intergenerational justice (one of the most consequential futures ethics concepts), value pluralism across cultures, and the ethics of existential risk. This theme deserves a much wider scope.

SOURCES ADDED

Nuffield Council on Bioethics (nuffieldbioethics.org)	Justification: THE UK authority on bioethics; produces landmark reports on gene editing, AI in healthcare, and emerging technology ethics; high academic credibility.

Ada Lovelace Institute (adalovelaceinstitute.org)	Justification: Leading data and AI ethics research institute; rigorous empirical and normative work on algorithmic fairness, facial recognition, and AI governance.

Partnership on AI (partnershiponai.org)	Justification: Multi-stakeholder body developing responsible AI norms; bridges academic, civil society, and industry perspectives on AI ethics.

Future Generations Commissioner (futuregenerations.wales.gov.uk)	Justification: Pioneering institutional model for intergenerational justice; directly relevant as a futures-oriented ethics institution.

IEEE Global Initiative (ethicsinaction.ieee.org)	Justification: Technical ethics standards body; produces Ethically Aligned Design — the leading framework for technology ethics standards.

FINAL SOURCE LIST (12 sources)

1	Oxford Martin School (oxfordmartin.ox.ac.uk) — futures ethics, future generations
2	OECD (oecd.org) — AI ethics principles
3	Nuffield Council on Bioethics (nuffieldbioethics.org) — bioethics
4	Ada Lovelace Institute (adalovelaceinstitute.org) — data and AI ethics
5	WEF (weforum.org) — global ethics, trust
6	Partnership on AI (partnershiponai.org) — responsible AI
7	Hawaii Research Center for Futures Studies (futures.hawaii.edu) — values futures

8	Pew Research Center (pewresearch.org) — social values, trust
9	IEEE Global Initiative (ethicsinaction.ieee.org) — technology ethics standards
10	Millennium Project (millennium-project.org) — ethical scenarios
11	CIFS (cifs.dk) — cultural values trends
12	Future Generations Commissioner (futuregenerations.wales.gov.uk) — intergenerational ethics

Theme 23: Society

FOCUS AREA EVALUATION

Current	*Inequality, social cohesion, civil society*
Revised	**Social inequality, polarisation and cohesion, social capital, civil society and social movements, community futures, loneliness and belonging, demographic social transformation**

Argumentation:

Adequate framing but underspecified for a futures platform. The description misses: social polarisation (politically critical), social capital erosion (Putnam tradition), community futures, loneliness as a public health and social crisis, the sociology of emerging technologies, and intersectionality in social analysis. "Civil society" is a broad descriptor that could be sharpened to "civil society and social movements." There is some overlap with Values & Ethics (22) and Health & Wellbeing (24).

SOURCES ADDED

UNDP (undp.org)	Justification: Publishes the Human Development Report — THE global benchmark for human development, inequality, and social progress; conspicuously absent.
Brookings Institution (brookings.edu)	Justification: Rigorous social policy research; authoritative on economic mobility, social capital, and inequality dynamics.
LSE (lse.ac.uk)	Justification: London School of Economics; premier social science institution with authoritative research on inequality, sociology, and social policy.
Joseph Rowntree Foundation (jrf.org.uk)	Justification: UK's leading poverty and inequality research foundation; produces authoritative annual UK Poverty report; relevant for evidence on inequality trends.
RSA (thersa.org)	Justification: Royal Society of Arts; produces innovative social analysis on belonging, community, and social innovation.

FINAL SOURCE LIST (12 sources)

1	OECD (oecd.org) — inequality, social well-being
2	UNDP (undp.org) — Human Development Report
3	WEF (weforum.org) — social cohesion, inclusion

4	Oxford Martin School (oxfordmartin.ox.ac.uk) — inequality, social change
5	Pew Research Center (pewresearch.org) — social trends
6	Brookings Institution (brookings.edu) — social policy
7	LSE (lse.ac.uk) — social science research
8	Chatham House (chathamhouse.org) — social and political change
9	SOIF (soif.org.uk) — social futures
10	RSA (thersa.org) — social innovation
11	Joseph Rowntree Foundation (jrf.org.uk) — poverty and inequality
12	Hawaii Research Center for Futures Studies (futures.hawaii.edu) — society futures

Category: People & Well-being

Theme 24: Health & Wellbeing

FOCUS AREA EVALUATION

Current	*Healthcare, mental health, biotech, longevity*
Revised	**Healthcare transformation, mental health futures, biotech and precision medicine, longevity and ageing health, pandemic preparedness, health equity, AI in healthcare, One Health**

Argumentation:

Good coverage but missing critical dimensions: pandemic preparedness (post-COVID urgency), precision and personalised medicine, AI in healthcare, health equity and the social determinants of health, and One Health (the interconnection of human, animal, and environmental health). The current source list of five is far too short — this is one of the most data-rich and institutionally supported futures themes.

SOURCES ADDED

Lancet (thelancet.com)	Justification: THE premier medical journal; publishes landmark health systems research, Lancet Commissions on key health futures issues (planetary health, climate, NCDs).
Wellcome Trust (wellcome.org)	Justification: World's largest independent biomedical research funder; publishes authoritative health research and policy analysis on global health challenges.
IHME (healthdata.org)	Justification: Institute for Health Metrics and Evaluation; produces the Global Burden of Disease — THE authoritative global health data resource.
Gates Foundation (gatesfoundation.org)	Justification: THE major global health funder; produces significant global health research on infectious disease, health equity, and vaccine futures.

RAND Corporation (rand.org)	Justification: Strong health policy research portfolio; rigorous health systems analysis and scenario planning.

ECDC (ecdc.europa.eu)	Justification: European Centre for Disease Prevention and Control; essential for European public health, pandemic preparedness, and health security.

NIH (nih.gov)	Justification: US National Institutes of Health; THE biomedical research authority; essential for biotech, precision medicine, and clinical research frontiers.

Nuffield Trust (nuffieldtrust.org.uk)	Justification: Leading health systems research institution; rigorous analysis of NHS futures, healthcare financing, and health workforce.

FINAL SOURCE LIST (13 sources)

1	WHO (who.int) — global health authority
2	Lancet (thelancet.com) — medical science, health systems
3	Wellcome Trust (wellcome.org) — health research
4	IHME (healthdata.org) — global disease burden data
5	Gates Foundation (gatesfoundation.org) — global health equity
6	WEF (weforum.org) — health transformation
7	Oxford Martin School (oxfordmartin.ox.ac.uk) — health futures
8	McKinsey (mckinsey.com) — healthcare economics
9	RAND Corporation (rand.org) — health policy
10	ECDC (ecdc.europa.eu) — European public health
11	NIH (nih.gov) — biomedical innovation
12	Nuffield Trust (nuffieldtrust.org.uk) — health systems
13	Nature (nature.com) — biomedical research

Theme 25: Education & Knowledge

FOCUS AREA EVALUATION

Current	*EdTech, lifelong learning, skills transformation*
Revised	**Education futures, lifelong and distributed learning, AI in education, skills transformation, future of universities, knowledge democracy, learning sciences**

Argumentation:

"EdTech" as a leading descriptor prioritises technology delivery over the fundamental transformations in what, how, and why we learn. The description misses: the future of universities and higher education credentials, AI in education (including academic integrity crises), knowledge

democracy and open access, learning sciences advances, and the geopolitics of knowledge production. "Skills transformation" should be framed as part of the broader knowledge economy transition.

SOURCES ADDED

UNESCO (en.unesco.org)	Justification: THE UN education authority; publishes Global Education Monitoring Report; indispensable for global education futures and education equity.
World Bank (worldbank.org)	Justification: Essential for education in developing countries, human capital development, and education financing — global comparative perspective.
KnowledgeWorks (knowledgeworks.org)	Justification: Specialist education futures organisation; produces authoritative foresight on learning and education system transformation.
EDUCAUSE Horizon Report (educause.edu)	Justification: Annual report on emerging technologies in higher education; THE standard reference for EdTech adoption and futures.
Stanford Graduate School of Education (ed.stanford.edu)	Justification: Leading academic institution for learning sciences, education innovation, and policy research.
WISE (wise.org.qa)	Justification: World Innovation Summit for Education; produces cross-cultural education innovation research and awards — important global perspective.

FINAL SOURCE LIST (12 sources)

1	OECD (oecd.org) — education systems, PISA, TALIS
2	UNESCO (en.unesco.org) — global education, GEM Report
3	World Bank (worldbank.org) — education development
4	WEF (weforum.org) — education and skills futures
5	European Commission (ec.europa.eu) — education skills agenda
6	IFTF (iftf.org) — learning futures
7	KnowledgeWorks (knowledgeworks.org) — education foresight
8	EDUCAUSE Horizon Report (educause.edu) — EdTech
9	Stanford GSE (ed.stanford.edu) — learning sciences
10	McKinsey (mckinsey.com) — education and skills
11	WISE (wise.org.qa) — education innovation
12	MIT OpenLearning (openlearning.mit.edu) — digital education

Theme 26: Work & Income

FOCUS AREA EVALUATION

Current	*Future of work, labour markets, UBI*
Revised	**Future of work, labour market transformation, automation and displacement, digital labour rights, income security models, skills transitions, labour geographies**

Argumentation:

"UBI" (Universal Basic Income) is a specific, contested policy proposal — listing it as a defining focus area gives it undue prominence and political colouring over other income security models (earned income tax credits, job guarantees, reduced working hours). The description should use "income security models" to remain appropriately agnostic. Missing: automation and technological displacement, the geography of job loss, gig economy rights, and skills transitions.

SOURCES ADDED

MIT Work of the Future (workofthefuture.mit.edu)	Justification: Rigorous interdisciplinary research on technology and work; one of the most authoritative academic programmes specifically on automation and employment futures.
Brookings Institution (brookings.edu)	Justification: Extensive future of work research, particularly on automation, geographic displacement, and workforce transitions.
Economic Policy Institute (epi.org)	Justification: THE authority on labour economics from a worker-centred perspective; essential for income inequality, wage dynamics, and union futures.
Oxford Internet Institute (oii.ox.ac.uk)	Justification: iLabour Project produces authoritative data on digital labour, gig platforms, and algorithmic management globally.
RSA (thersa.org)	Justification: Future of Work Centre produces innovative research on worker power, basic income experiments, and work meaning.
Fairwork Foundation (fair.work)	Justification: Empirical research on platform work conditions across 40+ countries; essential for gig labour rights dimension.

FINAL SOURCE LIST (12 sources)

1	ILO (ilo.org) — labour policy and standards
2	OECD (oecd.org) — future of work
3	WEF (weforum.org) — jobs and skills
4	McKinsey Global Institute (mckinsey.com) — automation impact
5	MIT Work of the Future (workofthefuture.mit.edu) — technology and work
6	Brookings Institution (brookings.edu) — employment futures
7	IFTF (iftf.org) — work futures
8	Oxford Internet Institute (oii.ox.ac.uk) — digital labour
9	Economic Policy Institute (epi.org) — labour economics

10	Pew Research Center (pewresearch.org) — work and employment attitudes
11	RSA (thersa.org) — future of work
12	Fairwork Foundation (fair.work) — platform work rights

Theme 27: Management & HR

FOCUS AREA EVALUATION

Current	*Leadership, org design, talent management*
Revised	**Leadership futures, organisational design transformation, talent management, distributed and hybrid work, DEI strategy, people analytics and HR technology, organisational culture**

Argumentation:

Adequate but misses critical contemporary dimensions: the transformation of organisational design under distributed/remote work (post-COVID structural shift), diversity, equity, and inclusion (DEI) as a strategic management imperative, the psychological safety framework, HR technology and people analytics, and organisational culture as competitive advantage. The theme could better distinguish itself from Theme 26 (Work & Income) by focusing on the management and organisational design level rather than the labour market level.

SOURCES ADDED

CIPD (cipd.org)	Justification: Chartered Institute of Personnel and Development; THE professional body for HR and people management; produces authoritative annual Good Work Index and HR research.
MIT Sloan Management Review (sloanreview.mit.edu)	Justification: Peer-reviewed management research with strong futures orientation; leadership, talent, and organisational design are core areas.
Gallup (gallup.com)	Justification: Produces the State of the Global Workplace report — THE benchmark for employee engagement, management effectiveness, and workplace culture data.
INSEAD (insead.edu)	Justification: Premier global business school; produces rigorous leadership and management research with genuinely international perspective.
Center for Creative Leadership (ccl.org)	Justification: Specialist leadership development research institution; produces authoritative data on leadership effectiveness and future leadership competencies.

FINAL SOURCE LIST (12 sources)

1	McKinsey (mckinsey.com) — leadership and management
2	HBR (hbr.org) — management innovation
3	Deloitte Human Capital (deloitte.com) — workforce trends
4	WEF (weforum.org) — talent and workforce

5	Gartner (gartner.com) — HR technology trends
6	CIPD (cipd.org) — people management
7	MIT Sloan Management Review (sloanreview.mit.edu) — management research
8	Gallup (gallup.com) — workforce engagement
9	INSEAD (insead.edu) — leadership research
10	Center for Creative Leadership (ccl.org) — leadership development
11	SHRM (shrm.org) — HR practices
12	Oxford Said Business School (ox.ac.uk) — management research

Theme 28: Leisure & Lifestyles

FOCUS AREA EVALUATION

Current	*Consumer culture, tourism, entertainment*
Revised	**Consumer culture transformation, experience economy, digital leisure and gaming, wellness and wellbeing economy, sustainable tourism, lifestyle futures, platform-mediated consumption**

Argumentation:

The description is accurate but insufficiently futures-oriented. Missing: the experience economy (Pine & Gilmore tradition), digital leisure and gaming (now larger than film and music combined), wellness economy, slow living and degrowth-aligned lifestyle trends, the sustainability of tourism, and platform-mediated consumer culture. "Consumer culture" as a concept should be interrogated rather than assumed — including both the commercialisation thesis and counter-movements.

SOURCES ADDED

UNWTO (unwto.org)	Justification: World Tourism Organization; THE authority on global tourism data, sustainable tourism policy, and tourism futures.
Euromonitor International (euromonitor.com)	Justification: Authoritative global consumer market data and trends; rigorous quantitative basis for consumer futures analysis.
JWT Intelligence / Wunderman Thompson (wundermanthompson.com)	Justification: Annual Future 100 trends report; one of the most cited consumer culture foresight publications globally.
Mintel (mintel.com)	Justification: Rigorous consumer market intelligence; produces authoritative lifestyle and consumer trend analysis.

FINAL SOURCE LIST (12 sources)

1	McKinsey (mckinsey.com) — consumer trends
2	OECD (oecd.org) — leisure, tourism economy
3	UNWTO (unwto.org) — tourism futures

4	WEF (weforum.org) — consumer lifestyle
5	Pew Research Center (pewresearch.org) — consumer and lifestyle trends
6	CIFS (cifs.dk) — lifestyle megatrends
7	Deloitte (deloitte.com) — consumer industry
8	IFTF (iftf.org) — consumer futures
9	Euromonitor (euromonitor.com) — global consumer markets
10	JWT Intelligence / Wunderman Thompson (wundermanthompson.com) — consumer culture trends
11	Mintel (mintel.com) — consumer insights
12	World Leisure Organization (worldleisure.org) — leisure futures

Category: Science & Demographics

Theme 29: Sciences & Research

FOCUS AREA EVALUATION

Current	*R&D, quantum computing, space, biotech research*
Revised	**Frontier sciences, convergent technologies, R&D systems transformation, open science, science-policy interface, quantum computing, space science, synthetic biology**

Argumentation:

The description cherry-picks specific technologies (quantum computing, space) rather than characterising the broader transformation of science and research. Missing: convergent technologies (nano-bio-info-cogno), open science and data sharing, the science-policy interface, transdisciplinary research, synthetic biology, and science ethics (dual-use research). Quantum computing and space are important but should not define the entire theme — they should appear as illustrative examples of frontier science. Biotech research overlaps with Theme 24 (Health & Wellbeing) and should be differentiated.

SOURCES REMOVED / REPLACED

Dubai Future Foundation (dubaifuture.gov.ae)	Reason: Government promotional body without scientific peer-review standards; insufficient for a Science & Research theme requiring high methodological rigour.

SOURCES ADDED

Science (science.org)	Justification: THE flagship journal of AAAS; one of the two most authoritative multidisciplinary science journals in the world; essential for this theme.
Royal Society (royalsociety.org)	Justification: UK's national academy of science; produces authoritative horizon scanning reports and policy documents on frontier science.
AAAS (aaas.org)	Justification: American Association for the Advancement of Science; largest general scientific society; essential for science policy and cross-disciplinary research futures.

European Research Council (erc.europa.eu)	Justification: THE European frontier research funder; publishes strategic analyses of emerging research fields and scientific priorities.

UNESCO (en.unesco.org)	Justification: UN science policy authority; publishes Science Report on global R&D trends; essential for open science governance and global science equity.

FINAL SOURCE LIST (13 sources)

1	Nature (nature.com) — scientific research authority
2	Science (science.org) — multidisciplinary research
3	Oxford Martin School (oxfordmartin.ox.ac.uk) — emerging science
4	Royal Society (royalsociety.org) — scientific leadership, horizon scanning
5	AAAS (aaas.org) — science advancement and policy
6	European Research Council (erc.europa.eu) — frontier research
7	OECD (oecd.org) — R&D indicators, science policy
8	Fraunhofer ISI (isi.fraunhofer.de) — applied research, innovation
9	IIASA (iiasa.ac.at) — systems science
10	CSER Cambridge (cser.ac.uk) — technology risk
11	PIK Potsdam (pik-potsdam.de) — earth system science
12	Future Earth (futureearth.org) — sustainability science
13	UNESCO (en.unesco.org) — science policy, open science

Theme 30: Population & Demographics

FOCUS AREA EVALUATION

Current	*Aging, migration, urbanisation trends*
Revised	**Population dynamics and fertility trends, ageing societies, migration and climate displacement, demographic dividends, labour demographics, generational transitions**

Argumentation:

"Urbanisation trends" creates direct overlap with Theme 10 (Construction & Urbanisation), which owns this territory. This theme should focus on the fundamental demographic drivers: population dynamics (fertility, mortality), ageing societies, migration and displacement (including climate migration — a major future driver), demographic dividends in the Global South, and the labour market consequences of demographic change. It should also include the critical interaction between demographics and AI — specifically population decline in advanced economies.

SOURCES ADDED

UN Population Division (population.un.org)	Justification: THE authoritative source for global population data, projections, and fertility/mortality analysis; produces World Population Prospects.

IOM (iom.int)	Justification: International Organization for Migration; THE authority on global migration data, policy, and forced displacement — essential for the migration dimension.

UNHCR (unhcr.org)	Justification: UN Refugee Agency; essential for displacement, refugee trends, and climate migration — increasingly a central demographic driver.

Population Reference Bureau (prb.org)	Justification: Authoritative population data and analysis; produces Population Bulletin — one of the most respected demographic publications.

Vienna Institute of Demography (oeaw.ac.at/vid)	Justification: One of Europe's leading demographic research centres; produces rigorous quantitative analysis of population trends and projections.

FINAL SOURCE LIST (13 sources)

1	UN Population Division (population.un.org) — World Population Prospects
2	OECD (oecd.org) — ageing societies
3	IOM (iom.int) — migration data and policy
4	UNHCR (unhcr.org) — displacement and refugees
5	Pew Research Center (pewresearch.org) — demographic trends
6	IIASA (iiasa.ac.at) — population scenarios
7	Oxford Martin School (oxfordmartin.ox.ac.uk) — demographics
8	Population Reference Bureau (prb.org) — population analysis
9	NIC (dni.gov) — demographic global trends
10	Brookings Institution (brookings.edu) — demographic change
11	WEF (weforum.org) — demographic futures
12	Vienna Institute of Demography (oeaw.ac.at/vid) — demographic research
13	CIFS (cifs.dk) — demographic megatrends

PART III: CROSS-CUTTING RECOMMENDATIONS

3.1 Source Governance Recommendations

The following principles should guide ongoing source curation for the 30 Futures Platform:

- Apply the tiered quality framework consistently. Tier 1 and Tier 2 sources should anchor each theme; Tier 3 and 4 sources should supplement rather than lead.
- Establish an annual source review cycle. Sources should be reviewed for institutional quality, continued activity, and relevance to the revised descriptions.
- Reduce commercial source overconcentration. McKinsey appears in 15+ themes; WEF in all 30. While both produce valuable analysis, their commercial interests (McKinsey) and stakeholder composition (WEF) warrant careful contextualisation.
- Increase Global South representation. Most current sources are Northern-hemisphere institutions. UNCTAD, UNDP, WFP, IOM, and CGIAR provide partially Global South perspectives, but regional think tanks (African Development Bank Institute, ECLAC, ADB Institute) could further enrich thematic coverage.
- Differentiate between monitoring sources and analytical sources. Some sources are primarily data providers (ITU, ILO, UN Population Division) while others are analytical institutions (Chatham House, Brookings). Both types are necessary; they serve different functions.

3.2 Description Design Principles

Based on the evaluation of all 30 theme descriptions, the following principles are recommended for future description updates:

- Avoid naming specific policy instruments (UBI, sanctions, carbon markets) as defining descriptors — these are debated tools, not themes. Use broader conceptual framings.
- Avoid anchoring descriptions in current technology names (big data, smart cities, IoT) that may date rapidly. Prefer process and transformation language.
- Ensure each description articulates a governance or power dimension alongside the technical dimension — futures studies is fundamentally about decision-making under uncertainty.
- Ensure descriptions differentiate adjacent themes clearly. The most confused adjacencies are: AI / Data; Business / Services; Climate / Sustainability; Health / Sciences. Each revised description now explicitly excludes what the adjacent theme covers.

3.3 Platform Architecture Recommendations

- Consider renaming Theme 15 from "Sustainability & Recycling" to "Sustainability & Circular Economy" — the current name significantly undersells the thematic scope.
- Consider splitting "Science & Demographics" into two categories to better reflect the epistemic distance between these domains.
- Introduce a cross-theme "horizontal layer" for themes that are genuinely pervasive: AI, climate change, and geopolitics shape virtually every other theme. Documenting these horizontal intersections would strengthen the platform's analytical coherence.
- Hybrid CoE (hybridcoe.fi — Helsinki) is recommended for Theme 21 (Security). As a Finland-based institution, it offers an opportunity to increase Nordic institutional representation in the platform while providing genuine specialist expertise on hybrid threats.

3.4 Summary Table: Description Changes

#	Theme	Current Description	Revised Description
1	**Artificial Intelligence**	*AI, machine learning, autonomous systems*	Generative AI, machine learning, autonomous systems, AI governance and regulation, AI safety, human-AI collaboration, societal and economic AI impacts

#	Theme	Current Description	Revised Description
2	**Data & Digitalisation**	*Big data, digital transformation, platforms*	Data governance and sovereignty, digital transformation, datafication, open data ecosystems, digital divide and inclusion, privacy and data rights
3	**Communication & Media**	*Telecom, journalism, social media, content*	Media transformation, journalism futures, platform governance, disinformation and synthetic media, social media dynamics, digital content ecosystems, trust in information
4	**Business**	*Business models, strategy, innovation management*	Business model transformation, corporate strategy, entrepreneurship and innovation ecosystems, organizational resilience, stakeholder capitalism, platform disruption
5	**Finance & Ownership**	*Banking, investment, fintech, ownership models*	Banking transformation, investment futures, fintech and CBDCs, financial inclusion, wealth distribution and inequality, alternative and commons-based ownership models
6	**Global Economy**	*Trade, macroeconomics, supply chains, sanctions*	Global trade, macroeconomics, supply chain transformation, geoeconomics, deglobalisation, emerging economies, sovereign debt, global inequality
7	**Digital & Sharing Economies**	*Platform economy, gig work, collaborative models*	Platform economy, algorithmic management and gig labour rights, peer-to-peer models, platform monopolies, collaborative and commons-based digital economies, digital labour governance
8	**Services**	*Service design, customer experience, servitization*	Service economy transformation, AI-augmented services, public service innovation, service design futures, platform-mediated services, knowledge-intensive services
9	**Industry & Manufacturing**	*Smart factories, automation, industrial IoT*	Smart manufacturing and Industry 5.0, industrial automation, industrial decarbonisation, reshoring and industrial resilience, additive manufacturing, green industrial policy
10	**Construction & Urbanisation**	*Smart cities, urban planning, building tech*	Urban futures and climate-resilient cities, housing systems, smart urban infrastructure, urban planning innovation, construction technology, built environment sustainability
11	**Energy**	*Renewables, grids, energy storage, nuclear*	Renewable energy transition, energy grids and storage, nuclear renaissance, hydrogen economy, energy security, just transition, energy geopolitics
12	**Freight & Logistics**	*Supply chains, last-mile, autonomous freight*	Supply chain resilience and transformation, freight decarbonisation, autonomous and digital logistics, nearshoring, maritime and port futures, last-mile innovation
13	**Private & Public Transport**	*Mobility-as-a-Service, EVs, public transit*	Electric vehicles and decarbonisation, autonomous mobility, Mobility-as-a-Service, shared and active mobility, public transit futures, aviation decarbonisation, mobility equity

#	Theme	Current Description	Revised Description
14	**Climate Change**	*Mitigation, adaptation, carbon markets*	Climate mitigation and adaptation, carbon markets and removal technologies, climate tipping points, climate finance and loss and damage, climate justice, geoengineering governance
15	**Sustainability & Recycling**	*Circular economy, waste, ESG reporting*	Circular economy, sustainable production and consumption, regenerative systems, ESG and corporate sustainability transformation, material flows, biodegradable innovation
16	**Nature**	*Biodiversity, ecosystems, land use*	Biodiversity loss and recovery, ecosystem services, natural capital valuation, nature-based solutions, rewilding, land and ocean governance, biodiversity finance
17	**Food**	*Agri-tech, food security, alternative proteins*	Food systems transformation, agricultural innovation, food security and sovereignty, alternative and cellular proteins, agroecology, food waste, nutrition futures
18	**Water & Oceans**	*Water scarcity, marine economy, blue tech*	Water security and freshwater scarcity, ocean governance and health, marine economy, ocean acidification, water-food-energy nexus, desalination and water technology
19	**Public Governance**	*Regulation, democracy, public-sector innovation*	Democratic governance futures, public-sector innovation, digital and AI governance, regulatory transformation, open government and citizen participation, institutional trust
20	**Geopolitics**	*International relations, conflicts, alliances*	Great power competition, multipolar world order, economic statecraft and tech nationalism, regional alliances, conflict dynamics, multilateral system transformation, climate geopolitics
21	**Security & Safety**	*Cybersecurity, defence, resilience, privacy*	Cybersecurity, hybrid warfare, critical infrastructure resilience, defence technology, biosecurity, information operations, privacy and surveillance, societal resilience
22	**Values & Ethics**	*Trust, transparency, corporate responsibility*	Digital and AI ethics, bioethics, intergenerational justice, social trust, value pluralism, corporate ethics and accountability, ethics of emerging technologies
23	**Society**	*Inequality, social cohesion, civil society*	Social inequality, polarisation and cohesion, social capital, civil society and social movements, community futures, loneliness and belonging, demographic social transformation
24	**Health & Wellbeing**	*Healthcare, mental health, biotech, longevity*	Healthcare transformation, mental health futures, biotech and precision medicine, longevity and ageing health, pandemic preparedness, health equity, AI in healthcare, One Health
25	**Education & Knowledge**	*EdTech, lifelong learning, skills transformation*	Education futures, lifelong and distributed learning, AI in education, skills transformation, future of universities, knowledge democracy, learning sciences

#	Theme	Current Description	Revised Description
26	**Work & Income**	*Future of work, labour markets, UBI*	Future of work, labour market transformation, automation and displacement, digital labour rights, income security models, skills transitions, labour geographies
27	**Management & HR**	*Leadership, org design, talent management*	Leadership futures, organisational design transformation, talent management, distributed and hybrid work, DEI strategy, people analytics and HR technology, organisational culture
28	**Leisure & Lifestyles**	*Consumer culture, tourism, entertainment*	Consumer culture transformation, experience economy, digital leisure and gaming, wellness and wellbeing economy, sustainable tourism, lifestyle futures, platform-mediated consumption
29	**Sciences & Research**	*R&D, quantum computing, space, biotech research*	Frontier sciences, convergent technologies, R&D systems transformation, open science, science-policy interface, quantum computing, space science, synthetic biology
30	**Population & Demographics**	*Aging, migration, urbanisation trends*	Population dynamics and fertility trends, ageing societies, migration and climate displacement, demographic dividends, labour demographics, generational transitions

CONCLUSION

The 30 Futures Platform Themes framework represents a substantively comprehensive and well-organised foundation for horizon scanning and strategic foresight. The six-category structure is coherent, the thematic scope is ambitious yet manageable, and the existing source base demonstrates awareness of major futures-relevant institutions.

The revisions recommended in this analysis aim to bring three improvements: (1) greater futures-orientation in theme descriptions, replacing technology-specific and policy-specific language with enduring conceptual framings; (2) higher source authority and independence, reducing reliance on commercial analysts and increasing the weight of intergovernmental, academic, and specialist institutions; and (3) sharper thematic differentiation, resolving overlaps between adjacent themes through clear definitional boundaries.

Following these revisions, each theme now carries a source list of 12–15 authoritative institutions that collectively span governance, scientific, civil society, and analytical perspectives. The resulting platform is better positioned to support rigorous, evidence-based futures analysis that transcends any single disciplinary or stakeholder perspective.

End of Report